

A7: Research field 7 - Extending the life of the built environment

1. Introduction, vision & mission

1.1 Introduction to the Research field

A substantial portion of the built environment has reached, or is approaching, the end of its original service life, with many structures no longer compliant with current design codes. In addition, evolving and unforeseen loading scenarios often lead to demands that exceed those assumed in design. Despite this, there is broad consensus that, in many cases, service life can be safely and cost-effectively extended through targeted, proactive interventions - provided that more advanced assessment tools and technologies are available. This research field addresses this need through coordinated efforts that integrate structural engineering, monitoring technologies, data science and AI, and reliability-based methods to understand, predict, and extend the service life of existing buildings and civil infrastructure.

1.2 Vision

RF7 envisions a built environment in which extending the service life of existing buildings and infrastructure becomes the preferred pathway. Through science-based, risk-informed, and sustainability-driven approaches, we enable safe, reliable, and resource-conscious performance, well beyond original design expectations.

1.3 Mission

RF7 develops an interdisciplinary framework for lifetime extension by quantifying deterioration and capacity, forecasting climate-driven demands, creating adaptive retrofit solutions, and delivering transparent risk-and-reliability-based sustainability-informed decision support that integrates structural health monitoring information.

2. State of the field

2.1 Global centres of excellence

Structural Reliability & Risk: RSUQ group, ETH Zurich, (<https://sudret.ibk.ethz.ch/>); ERA Group, TU Munich (<https://www.cee.ed.tum.de/era/era-group/>); SURI group, Stanford University (<https://urbanresilience.stanford.edu/>); Reliable Structures Department, Mobility & Built Environment Unit, TNO (<https://www.tno.nl/en/about-tno/organisation/units/mobility-built-environment/reliable-structures/>); TU Delft (<https://www.tudelft.nl/en/ceg/about-faculty/departments/hydraulic-engineering/sections/hydraulic-structures-and-flood-risk/>)

The Risk, Safety & Uncertainty Quantification (RSUQ) Group at ETH Zurich, led by Prof. Bruno Sudret, is internationally recognised for advancing probabilistic methods in structural engineering. At Technical University of Munich (TUM), the Engineering Risk Analysis Group, founded by Prof. Daniel Straub, is widely known for its contributions to probabilistic modelling of infrastructure and natural hazards. The group develops Bayesian networks, system reliability methods, and risk-based decision frameworks. The Stanford Urban Resilience Initiative (SURI) at Stanford University brings a multi-disciplinary perspective to structural reliability, focusing on urban systems subjected to natural hazards such as earthquakes, floods, and wildfires. Their research integrates structural reliability modelling with socio-technical systems analysis, emphasising performance-based design, community-level risk assessment, and resilience optimisation. The Reliable Structures Department at TNO (Mobility & Built Environment Unit) is a leading applied research centre in Europe for performance- and reliability-based assessment of civil and offshore structures. TNO specialises in integrating structural reliability theory with full-scale testing, monitoring data, and industry-driven applications. At TU Delft, research within the Hydraulic Structures and Flood Risk section addresses reliability and risk in the context of hydraulic infrastructure, flood defences, and coastal systems. Their work strongly influences Dutch and international practice in flood-risk management, integrating structural reliability with climate adaptation strategies.

Wind Engineering & Climate-Loading: BLWT laboratory, Western University Canada (<https://www.uwo.ca/projects/heritage/heritage4/>); WERC, Tokyo Polytechnic University (<https://www.collaborate.wind.t-kougei.ac.jp/en/>); CWE, RWTH Aachen University (<https://www.cwe.rwth-aachen.de/home-2/>).

Leading centres such as the Boundary Layer Wind Tunnel Laboratory (BLWTL) at Western University, the Wind Engineering Research Center (WERC) at Tokyo Polytechnic University, and the Institute for Structural Mechanics and Wind Engineering (CWE) at RWTH Aachen University represent the forefront of wind engineering and climate loading research. At Western University, Professor Greg Kopp has been a leading figure in advancing boundary layer wind tunnel testing, full-scale measurements, and wind load modelling on buildings and bridges, contributing to improved design standards and resilience against extreme wind events. At Tokyo Polytechnic University, Professor Yukio Tamura has made seminal contributions to wind-resistant design, aerodynamic stability, and wind-induced vibration of structures, shaping international standards and advancing understanding of typhoon and strong wind effects on the built environment. At RWTH Aachen University, Professor Christian Bucher and Professor Christoph Geißler have contributed to probabilistic modelling, stochastic dynamics, and experimental wind engineering, focusing on the interaction between wind loads and structural response under uncertainty. Collectively, these centres integrate experimental, numerical, and probabilistic approaches to quantify wind and climate-induced loads, supporting the design and assessment of resilient infrastructure under changing environmental conditions.

Structural Health Monitoring & Digital Twins: V4RL/IRIS labs, ETH Zurich (<https://v4rl.com/>); SMM group, ETH Zurich (<https://chatzi.ibk.ethz.ch/>); Dynamics Research Group, University of Sheffield (<https://drq.ac.uk/>); STAIRlab, UC Berkeley (<https://stairlab.berkeley.edu/>) Civil Engineering Safety and Durability, Tsinghua University.

(<https://www.civil.tsinghua.edu.cn/ceen/info/1052/1341.htm>); Smart Structures Technology Laboratory (SSTL), University of Illinois Urbana (<https://sstl.cee.illinois.edu/>); Illinois Structural Health Monitoring Project (ISHMP) (<http://shm.cs.illinois.edu/>).

Leading international centres such as ETH Zurich, the University of Sheffield, UC Berkeley, Tsinghua University, and the University of Illinois Urbana-Champaign are driven by key researchers who have shaped the modern field of structural health monitoring and digital twins. At ETH Zurich, Professor Eleni Chatzi has made seminal contributions to physics-informed machine learning, system identification, and digital twinning, developing hybrid modelling frameworks that fuse real-time data with reduced-order physics-based models for infrastructure monitoring and decision support. At the University of Illinois, Professor Bill Spencer has been instrumental in advancing wireless smart sensor networks and decentralized monitoring systems, enabling scalable, real-time SHM deployments on full-scale infrastructure and establishing new paradigms for autonomous monitoring. At the University of Sheffield, researchers such as Professor Keith Worden and Professor Elizabeth Cross have pioneered data-centric and machine learning approaches to SHM, including novelty detection, population-based SHM, and uncertainty quantification, which are central to modern digital twin concepts. At UC Berkeley, Professor Kenichi Soga and colleagues at STAIRlab have driven innovations in distributed sensing, cyber-physical systems, and data integration for infrastructure monitoring, with strong emphasis on field deployment and infrastructure resilience. At Tsinghua University, leading researchers in civil engineering safety and durability have advanced large-scale monitoring systems and durability assessment methods for bridges and major infrastructure, combining sensing technologies with long-term performance evaluation under environmental and operational loading.

<https://www.ucl.ac.uk/infrastructure-systems>/<https://www.tudelft.nl/citg/over-faculteit/afdelingen/materials-mechanics-management-design-3md/integral-design-management/groups/engineering-systems/research><https://www.ntnu.edu/ivb/civil-engineering-and-geomatics>

Adaptive Systems & Retrofitting: SSTL, University of Illinois (<https://sstl.cee.illinois.edu/>); SMARTI Lab, University of Illinois (<https://sychterz.cee.illinois.edu/>); University of Stuttgart (<https://www.sfb1244.uni-stuttgart.de/en/>); Space Structures Laboratory, Caltech (<https://www.pellegrino.caltech.edu/>); Compliant Mechanisms, TU Chemnitz (https://www.tu-chemnitz.de/mb/mp/forschung/nachgiebige_systeme/index.html.en)

Leading centres such as the Smart Structures Technology Laboratory (SSTL) and SMARTI Lab at the University of Illinois Urbana-Champaign, the Collaborative Research Centre SFB 1244 at the University of Stuttgart, the Space Structures Laboratory at Caltech, and the Compliant Mechanisms group at TU Chemnitz represent the forefront of adaptive systems and structural retrofitting. At the University of Illinois, Professor Bill Spencer has been a central figure in the development of smart damping and semi-active control systems, including magnetorheological devices, enabling real-time structural adaptation and advanced retrofit solutions for civil infrastructure. At the University of Stuttgart, Professor Werner Sobek has led pioneering work on adaptive and lightweight structures, developing systems that actively respond to changing loads and environmental conditions to improve resource efficiency and extend structural lifespan. At Caltech,

Professor Sergio Pellegrino has advanced deployable and transformable structures, including origami-inspired systems and lightweight adaptive mechanisms, with strong relevance for reconfigurable and retrofittable structural concepts. At TU Chemnitz, Professor Lutz G. Friebe and colleagues have contributed significantly to the development of compliant mechanisms, enabling flexible, jointless systems that allow controlled motion and adaptability, with potential applications in structural components and retrofit technologies. Together, these centers integrate control systems, material innovation, and structural mechanics to enable adaptive, resilient, and resource-efficient solutions for both new and existing infrastructure.

Structural Engineering and Modelling: ETH Zurich, Concrete Structures and Bridge Design (<https://kaufmann.ibk.ethz.ch/>); ETH Zurich, Steel and Composite Structures (<https://taras.ibk.ethz.ch/>); Structural Engineering Research Group, Universidad Politécnica de Madrid (<https://www1.caminos.upm.es/estructural/language/en/>).

ETH Zurich holds a leading international position in structural engineering, particularly in the fields of concrete structures and steel structures. The Chair of Concrete Structures and Bridge Design, led by Prof. Walter Kaufmann, is widely recognised for its rigorous development of mechanically consistent models for the design of new and assessment of existing concrete structures. The Chair of Steel and Composite Structures, headed by Prof. Andreas Taras, is well-known for innovative, industry-oriented research, including models for fatigue capacity of steel structures and performance-based fire design. Both chairs have world-class experimental facilities, enabling validation through large-scale testing. At Universidad Politécnica de Madrid (the Structural Engineering Research Group) Prof. Miguel Fernández Ruiz (formerly at *EPFL*) continues his internationally recognised work on capacity models for reinforced concrete. He has contributed to research that forms the basis of the shear and punching shear provisions in the next generation of Eurocode 2.

Deterioration modelling: International Center for Numerical Methods in Engineering (CIMNE), Structural Mechanics (<https://cimne.com/research/research-clusters/structural-and-particle-mechanics/>); University of Texas at Austin (UTA) <https://sites.utexas.edu/ferche/> and The Laboratory of Materials and Durability of Constructions (LMDC), Toulouse <https://www.lab-lmdc.fr/en/le-laboratoire-english/#>, University of Ottawa <https://www.uottawa.ca/faculty-engineering/departement-civil-engineering>

CIMNE holds a leading international position in the modelling of corrosion-induced deterioration of reinforced concrete. The group's work is strongly influenced by Prof Carmen Andrade, one of the leading experts on reinforcement corrosion mechanisms, monitoring and predictive modelling. University of Texas at Austin is together with University of Ottawa, some of the leading institutions on structural capacity of ASR deteriorated concrete structures. Their contributions help quantify the long-term structural implications of ASR damage and support the development of assessment and mitigation strategies for affected infrastructure. Many other research groups have focused on the material and chemical understanding of ASR deterioration mechanism. Among those, the Laboratory of Materials and Durability of Constructions (LMDC) in Toulouse is widely recognized for its mechanistic studies and predictive models related to ASR expansive chemical reactions. LMDC

integrates microstructural analysis, chemo-mechanical modelling, and long-term exposure studies to characterize ASR-induced cracking.

Benefits for Danish Research through Collaboration

Collaboration with these leading international centres would directly support the objectives of the CEBE programme, in particular within the research fields related to lifetime extension of structures, climate resilience, and sustainable building and infrastructure systems. Access to state-of-the-art methodologies in structural reliability, capacity modelling, steel fatigue, wind and climate loading, structural health monitoring, digital twins, and adaptive systems would strengthen the scientific foundation of Danish research and enhance the ability to address complex, multi scale challenges across the built environment.

Such partnerships would enable integration of advanced sensing, physics informed modelling, and probabilistic risk frameworks into Danish research on both buildings and infrastructure, supporting more accurate condition assessment, predictive maintenance, and decision making under uncertainty. Access to large scale experimental facilities, long term monitoring datasets, and validated digital twin platforms would accelerate the transition from research to implementation, which is central to the CEBE ambition of bridging fundamental research and real-world application.

Furthermore, collaboration would enhance the development of resource efficient and adaptive design strategies for both buildings and infrastructure, contributing to reduced material use, extended service life, and improved resilience of the built environment within planetary boundaries. It would also strengthen Denmark's international research positioning, support joint funding initiatives, and foster the training of early career researchers within a globally connected research ecosystem.

2.2 State of the art and DK position in the research field

RF7.1 Assessing the capacity and residual life of buildings and civil structures

Assessment of load-carrying capacity of aging structures is still largely governed by deterministic code-based checks originally developed for new design. This is suboptimal for non-code compliant and deteriorated structures, where material degradation and evolving loading conditions play a significant role. While key components for advanced assessment exist - such as monitoring techniques (e.g. ETH Zürich, TUM, TU Delft), load-capacity models (EPFL, UPM, ETH Zürich, DTU, SDU), deterioration modelling including corrosion and ASR (CIMNE, UPM, DTU, Chalmers, UTA), and probabilistic decision frameworks (Stanford, University of Colorado, Trinity College Dublin) - these elements remain fragmented and lack a unified, validated framework with consistent interaction between them.

Structural health monitoring (SHM) has matured significantly, encompassing sensor fusion, operational modal analysis, vision-based systems, UAV and robotic inspections, and virtual sensing approaches. However, in current practice, monitoring data is rarely translated into quantitative updates of structural capacity or reliability. The integration of SHM with physics-based models

through Bayesian updating, Kalman filtering, and inverse methods remains limited, and end-to-end frameworks linking measurement, model updating, and decision-making are still lacking.

In Denmark, a strong combination of modelling expertise, monitoring capabilities, and experimental infrastructure provides a unique foundation for addressing this gap. Research spans transport mechanisms (Sharmilan et al., 2024), corrosion propagation modelling (Michel et al., 2016), ASR-induced degradation (Hansen, 2019; Barbosa et al., 2018), and mechanical shear-transfer models for non-code compliant structures (Autrup et al., 2023; Sørensen et al., 2025), supported by efficient numerical tools (Vestergaard et al., 2021).

At the same time, Denmark has a strong tradition in vibration-based SHM and operational modal analysis, with increasing focus on data assimilation techniques such as Kalman filtering for system identification, load estimation, and damage detection. Recent developments include applications in weigh-in-motion and model updating (Lai et al., 2022), physics-informed latent force models for complex loading (Caglio et al., 2025), and machine-learning-assisted fatigue prognosis using multi-fidelity modelling (Al-Hagri et al., 2026). Emerging technologies such as UAV- and robotics-based inspections (Kapoor et al., 2021) further expand the data acquisition capabilities, although their integration into assessment frameworks remains limited.

RF7.1 will bridge these gaps by developing a **coherent, data-informed assessment framework**, where monitoring data, deterioration models, and structural analysis are consistently integrated through probabilistic updating. This will enable condition-dependent evaluation of residual capacity and remaining service life, moving beyond deterministic checks toward continuously updated, reliability-based assessment of existing structures.

RF7.2 Predicting future use and load scenarios of buildings and civil structures

State-of-the-art load scenario prediction increasingly combines wind engineering, climate projections, and machine learning, including inverse methods for inferring loads from structural response (Zhi et al., 2017). However, engineering practice still relies largely on stationary load assumptions, and current codes do not adequately account for evolving climate-driven phenomena such as non-synoptic storms.

Micro- and mesoscale forecasting capabilities are rarely integrated into reliability assessments or consequence-based decision models, leading to fragmented lifetime extension strategies. While climate storyline approaches have begun to quantify intensification of extreme events (Øelund et al., 2026), these are not yet systematically incorporated into structural workflows.

In Denmark, research is advancing toward data-driven predictive frameworks, including LSTM-based short-term extreme wind forecasting (Mode et al., 2025) and exploration of directionality–predictability relationships (Modé et al., 2024).

RF7.3 Engineering solutions for life extension of buildings and civil structures

Engineering solutions for life extension increasingly draw on advances in structural aerodynamics and control. Concepts such as flexible retrofit configurations, aerodynamic modifications, and

vortex-induced vibration mitigation build on established flow–structure interaction principles (Andersen & Brandt, 2018; Gao et al., 2021).

However, current approaches face limitations in scalability, applicability across structural typologies, and practical constraints related to retrofit implementation. Most solutions remain based on deterministic loading and classical control strategies, with limited incorporation of data-driven and probabilistic adaptation, and scarce full-scale validation.

Recent developments in Denmark include adaptive aerodynamic control strategies, spanning compliant mechanism synthesis, selective compliance, shape morphing, and model predictive vibration control (Seltmann & Hasse, 2025; Reksowardojo et al., 2022; Dakova et al., 2025).

RF7.4 Risk-based decision support systems for extending the life of the built environment

Risk- and reliability-based approaches are increasingly recognised as valuable tools for decision making under uncertainty; particularly in light of existing structures and the green transition (Köhler and Baravalle, 2019; Diamantidis et al., 2025; Melhem et al., 2020; Hingorani and Köhler, 2023). As a result, the shift towards these methods is also increasingly reflected in standards and guidance documents, see e.g. EN 1990:2023 or DS 11990:2024. Nevertheless, their routine application in engineering practice remains limited. A key reason is the lack of concrete and consistent guidance for the many (probabilistic) modelling choices required in such assessments. Consequently, there is an increasing call for clearer and more transparent practical guidance on probabilistic modelling choices (Hingorani and Köhler, 2023; Diamantidis et al., 2025; Steenbergen and Vrouwenvelder, 2025).

Denmark has a long-standing tradition of risk- and reliability-based methods. Reliability-based code calibration has been part of Danish practice for decades (Sørensen, 2001). This is also reflected in national guidance for reliability-based assessment of existing bridges issued by the Danish Road Directorate (Vejdirektoratet, 2024). In addition, risk-based decision making has been widely explored in areas such as wind turbines (Nielsen and Sørensen, 2011). Ongoing work also addresses challenges related to probabilistic modelling, including system reliability modelling (Meinen and Steenbergen, 2025) and model uncertainties in wave load models (Meinen et al., 2020).

3. How will the research field address the core challenges?

RF7 addresses one of the core challenges of CEBE by shifting engineering practice from replacement toward active, risk-informed lifetime extension of existing infrastructure.

The fundamental challenge is not the absence of knowledge, but the fragmentation between structural assessment, monitoring, environmental loading, and decision-making. RF7 resolves this by developing a coherent, interdisciplinary framework that integrates structural engineering, data science, and reliability-based methods into a unified workflow.

First, RF7 establishes quantitative models for deterioration and residual capacity, capturing time-dependent mechanisms such as corrosion propagation, fatigue crack growth, and material

degradation within probabilistic frameworks. These models will be integrated with advanced numerical methods and experimental validation to enable condition-dependent capacity assessment of non-code-compliant structures, reducing epistemic uncertainty and avoiding overly conservative deterministic checks.

Second, RF7 advances data-driven structural assessment by transforming monitoring data into actionable information. Through structural health monitoring, UAV- and robotic-based inspections, and virtual sensing, combined with physics-informed machine learning and Bayesian data assimilation, structural states, loads, and model parameters will be continuously updated. This enables near real-time system identification, eventual damage detection, and probabilistic model updating, leveraging CEBE's cross-cutting enablers in digitalization and automation.

Third, RF7 develops and evaluates adaptive and retrofit solutions that enhance structural performance and extend service life. This includes both conventional strengthening approaches and advanced concepts such as aerodynamic load mitigation, morphing structural components, and adaptive or performance-enhancing intervention strategies. These interventions are formulated within reliability-based design and assessment frameworks to ensure safety while minimizing material use and intervention, supporting circularity and design for regeneration principles.

Fourth, the research establishes probabilistic frameworks for future loading and climate effects, linking large-scale climate projections with local structural demand through multi-scale modelling and machine learning. Inverse methods and system identification techniques will enable estimation of environmental loads from measured structural response, while accounting for non-stationary processes and extreme events. This directly supports the CEBE objective of climate-resilient infrastructure, ensuring that lifetime extension strategies remain robust under evolving environmental conditions.

Finally, RF7 will strive to develop AI-supported, risk-informed decision-support tools that integrate monitoring data, deterioration modelling, load prediction, and reliability analysis into transparent engineering workflows. These tools will enable quantitative evaluation of intervention strategies (repair, retrofit, or replacement) under uncertainty, supporting lifecycle optimization and sustainability assessment while reducing CO₂ emissions associated with reconstruction.

By integrating these components, RF7 acts as a system-level enabler across CEBE, linking digitalization, sustainability, and climate adaptation into implementable engineering solutions. The outcome is a transition from conservative, fragmented assessment practices toward data-driven, reliability-based, and resource-efficient lifetime extension of the built environment.

3.1 Global Perspectives

The research within this field is primarily focused on challenges relevant to lifetime extension of ageing building and infrastructure systems, subjected to changing climate and increasing loading. These challenges are particularly pronounced in developed economies, where the built environment is older and subject to stringent performance, safety, and durability requirements. In this highly industrialized context, the methodologies being developed, including structural health

monitoring, digital twins, probabilistic risk assessment, and adaptive systems, are very likely to be quickly adopted by both asset managers and the industry.

As building stocks and infrastructure systems in the Global South continue to expand and mature, they will increasingly face similar challenges related to durability, maintenance, and climate resilience in the future. Moreover, due to underfunded maintenance, such scenarios will be likely to occur earlier than in developed economies. Therefore, the research conducted in this research field is very relevant and timely world-wide.

However, a major obstacle to infrastructure lifetime extension in the Global South is the limited availability of engineering data. Many regions lack complete infrastructure inventories, systematic inspection records, maintenance histories, and reliable traffic or environmental monitoring data. As a result, sophisticated AI-based methods, which require extensive digitalization and large amounts of data might not be applicable soon. Instead, it is envisioned that low-data tools such as the risk-based decision framework developed in RF7.4 will play a central role in projecting CEBE's impact there. By enabling risk-based decision-making, decisions can be based on whatever data is available, making the approach robust even in data-scarce contexts. In regions with limited data availability, the resulting higher uncertainties are explicitly incorporated into the risk analysis, ensuring that decisions remain transparent and systematically informed despite data constraints.

A key pathway for global impact is through contributions to international codes of practice and standardisation bodies, including fib, PTI, IABSE, and JCSS. Through active engagement in these organisations, the research is expected to influence future design guidelines, reliability frameworks, and lifecycle management practices for both buildings and infrastructure. This approach ensures immediate relevance within advanced economies while enabling long term international impact through the gradual diffusion of knowledge into global engineering practice, without diluting the scientific and technical focus of the research field.

4. Impact framework

4.1 Intended impact

Once the research outcomes are implemented, society will increasingly maintain, adapt, and extend the service life of existing buildings, bridges, and other civil infrastructures rather than replacing them. A larger share of the built environment will undergo transformation and structural adaptation that safely prolongs its use, resulting in fewer demolitions and fewer new constructions. Better knowledge of deterioration processes and more reliable assessment of the actual structural capacity of infrastructures, including their performance in fire, will reduce uncertainty about the current and future condition. This will allow authorities and owners to plan retrofitting, repair, or delayed replacement in a more informed and strategic manner. Therefore, maintenance actions that require large amounts of raw materials and energy will be avoided more often, leading to a substantial reduction in CO₂ emissions. Communities will also benefit from safer infrastructures, reduced costs related to repair, or reconstruction, and improved decision making regarding structural adaptation. Over time, this will lead to broader adoption of adaptive

engineering solutions that enable safe lifetime extension of infrastructures while supporting sustainability and long-term resilience of the built environment.

4.2 Contextual conditions

Barriers

- Engineering practice relies on established codes and conservative design traditions, which can slow acceptance of new assessment approaches for existing structures.
- Limited long term monitoring data and uncertainty about deterioration processes may reduce confidence in decisions to extend service life.
- Implementation requires new analytical competencies and digital tools, which may initially challenge consulting firms and infrastructure owners.
- Regulatory approval procedures and liability concerns can discourage stakeholders from applying innovative structural assessment and retrofit strategies.
- Fragmented ownership and management of infrastructure assets may complicate coordinated adoption of new maintenance and assessment frameworks.

Facilitation is intended to happen through university/industry courses, code-committee activity, and industrial pilot projects.

Opportunities

- Aging infrastructure networks create strong demand for reliable methods that safely extend structural service life.
- Climate targets and resource efficiency policies encourage solutions that reduce demolition, reconstruction, and associated material and energy consumption.
- Advances in sensing technologies, data analysis, and modelling enable improved monitoring, reliable diagnosis and prognosis of infrastructure performance and health.
- Collaboration between universities, consulting firms, and public authorities supports rapid transfer of validated research results into engineering practice.
- Greater awareness of lifecycle sustainability encourages decision making that prioritizes adaptation and transformation of existing buildings and infrastructures.

4.3 CEBE impact levers

- Develop practical guidelines, demonstration projects, and open datasets to translate validated models and monitoring methods into engineering workflows used by consultants, infrastructure owners, and public authorities.
- Establish an **RF7 Industry Advisory Board** comprising representatives from infrastructure owners, consulting engineering firms, contractors, technology providers, and public authorities. The board will provide strategic feedback on research priorities, identify barriers to implementation, and help ensure that research outputs address real industry needs and opportunities.
- Collaborate with consulting firms, infrastructure owners, and public authorities to co-develop and validate methods through pilot projects and demonstration cases. Industry

experts will be involved where appropriate in project steering groups, co-supervision of MSc, PhD and postdoctoral researchers, guest lectures, workshops, and examination activities. Examples may include collaboration with the Danish Road Directorate, which manages critical bridge assets within the national transport network, while facilities scheduled for transformation or decommissioning, such as parts of the existing Odense University Hospital campus, may provide unique testbeds for validating new monitoring technologies, assessment methods, and decision-support frameworks. The strong industrial networks of SDU, AU, and DTU, including collaboration with engineering consultancies such as COWI, Ramboll, and NIRAS, will help ensure continuous alignment between academic developments and engineering practice.

- Establish pilot monitoring platforms and data-sharing partnerships among universities, industry, and public authorities to address data gaps, accelerate technology validation, and support coordinated infrastructure management.
- Utilise the strong position of CEBE to secure external funding for follow-up research, innovation, and demonstration activities, including industrial PhDs, industrial postdoctoral projects, collaborative research initiatives, and implementation-oriented projects with industry partners.
- Support innovation and knowledge transfer through dedicated innovation management resources and close engagement with industry. Where appropriate, research outcomes may provide the foundation for new startups, spin-outs, software tools, digital services, and monitoring technologies that can accelerate adoption and societal impact.
- Engage regulators, standardisation bodies, and professional organisations through established links to organisations such as CEN, IABSE, fib, and JCSS to facilitate the integration of new reliability frameworks, assessment methodologies, and lifetime-extension approaches into engineering standards, guidelines, and approval procedures.
- Provide training programmes, workshops, continuing education activities, and digital tools to support the adoption of advanced modelling, monitoring, AI-enabled inspection, and data-driven decision-support methods by practicing engineers and infrastructure managers.
- Update MSc programmes and educational activities to equip future engineers with the competencies required to assess, maintain, retrofit, and extend the life of the built environment using the methods and technologies developed within RF7.

Intended outcomes (and main beneficiary / stakeholder)

To achieve the desired impact, the methods and frameworks developed in the research field must be widely adopted by professional engineers, infrastructure owners, and public authorities responsible for managing the built environment. Engineering consulting firms play a central role since they will apply the new scientific knowledge and analytical tools in practical assessments, monitoring activities, and maintenance planning for bridges, buildings, and other civil infrastructures. By adopting these approaches, consultants will be able to evaluate structural performance more reliably, monitor larger portfolios of infrastructure assets, and support safer and longer service lives through informed maintenance and retrofit strategies. Infrastructure owners and operators, including public agencies, municipalities and private foundations, will benefit from improved decision support for planning repairs, retrofitting, or delayed replacement. The research

will also strengthen the capacity of consulting firms to quantify environmental actions such as wind loading through virtual sensing and predictive environmental modelling. These advances will increase awareness and acceptance of adaptive structural systems as viable solutions for extending service life. At the same time, the generated knowledge will inform future discussions on reliability targets in structural design and assessment standards. The main beneficiaries will therefore include infrastructure owners, engineering consulting firms, public authorities, and the wider community that depends on safe, reliable, and sustainable infrastructure.

4.4 Research outputs

The research field will deliver a set of validated scientific methods, digital tools, experimental datasets, and practical frameworks that enable reliable assessment and lifetime extension of the existing built environment.

To support lifetime extension of the existing build environment in general, RF7 will develop sensing and monitoring technologies that support data driven assessment and adaptive operation of infrastructure. The work will deliver systems for automated inspections using unmanned aerial vehicles, algorithms for virtual sensing of wind loading from structural response measurements collected through multi modal sensor networks, and artificial intelligence methods that support the assisted operation of structural health monitoring systems. The research will also produce synthesis and control approaches for adaptive structural systems based on formal optimization and data driven modelling, enabling infrastructures to respond to changing environmental and operational conditions while extending their service life. Together, these outputs will form a comprehensive scientific and technological foundation for assessing, monitoring, and extending the lifetime of the built environment.

For assessment and lifetime extension of existing concrete structures, RF7 will produce more accurate and computationally efficient methods for assessing the load bearing capacity of reinforced concrete bridges and buildings that do not comply with current design standards (e.g. lack of minimum reinforcement and/or use of outdated reinforcement type and layout), supported by experimental data for validation. This work will also provide improved modelling tools and experimental techniques for characterizing rebar corrosion and alkali silica reaction and for understanding how these deterioration mechanisms influence structural behaviour.

For assessment and lifetime extension of existing steel structures, RF7 will deliver a framework for diagnosing fatigue induced cracking and estimating the remaining fatigue life of steel structures, supported by multi fidelity structural models, virtual sensing tools based on Kalman filtering for estimating loads and stresses, and advanced fracture mechanics-based fatigue models for simulating crack initiation and growth.

For more accurate assessments of load- and environmental impact, RF7 will develop machine learning models for probabilistic prediction of wind conditions affecting structural performance, and a multi physics modelling framework that links large scale climate developments with the local environmental conditions acting on structures.

To ensure optimized and safe lifetime extension of buildings and bridges, RF7 provides practical improvements to current risk- and reliability-based assessment frameworks, which can incorporate the models and experimental knowledge established in RF7. Where useful, improved guidance on target reliabilities for existing structures is established.

In addition to above specific output, RF7 has the ambition to acquire additional funding for research that will focus on:

- Evaluation procedures and numerical methods for assessing and extending infrastructure resilience to modern (and future) fire loads.
- Finite Element Limit Analysis of existing masonry structures
- Extended pilot studies for advanced structural health monitoring systems, with a focus on damage detection.
- Extension of extreme event analysis using combined physical modelling and data-driven techniques.
- Use of aerodynamic control as a means of active excitation for condition assessment.
- A digital engineering tool for risk- and reliability-based decision making that is applicable for practicing engineers without specialised probabilistic expertise. The tool should embed the relevant probabilistic modelling choices and methods, be able to incorporate in-situ data, and be based on both newly developed and existing probabilistic modelling guidance.

4.5 Research activities

The first group of activities focuses on assessing the capacity and residual life of structures. This includes the development of advanced analytical and numerical approaches for evaluating the load bearing capacity of reinforced concrete bridges and buildings that do not comply with current design provisions. The research will also include modelling and laboratory investigations of deterioration mechanisms such as rebar corrosion and alkali silica reaction and their influence on structural behaviour and safety. In parallel, methods will be developed for diagnosing and predicting fatigue damage in steel structures through multi-fidelity fracture mechanics models supported by ML algorithms as well as virtual sensing of loads and stresses. Field based inspection and monitoring technologies will also be investigated, including automated structural inspections using unmanned aerial vehicles combined with computer vision and artificial intelligence supported structural health monitoring systems.

A second group of activities addresses the prediction of future loading and use scenarios for buildings and infrastructures. Research will develop software sensing approaches that infer wind loading from structural response data and machine learning models that support probabilistic prediction of wind conditions affecting structural performance. These developments will be combined with analyses of how climate change may influence environmental loading processes and how these changes affect structural safety and long-term durability.

A third group of activities focuses on engineering solutions that enable safe lifetime extension. Research will investigate strengthening and repair strategies that increase the capacity and remaining service life of buildings and civil infrastructures. The work will also explore

adaptive solutions such as active aerodynamic control of slender or line-like structures to reduce loading effects and improve structural resilience.

A final group of activities will develop decision support frameworks for risk-informed lifetime extension of the built environment. This includes determining suitable reliability levels for existing structures, establishing improved probabilistic assessment frameworks for standardized evaluations, and developing practical guidance for risk informed decisions on repair, retrofitting, and transformation. Data driven modelling approaches will also be investigated to support the planning and design of retrofitting and structural transformation processes. Together, these activities will provide the scientific and technological basis needed to assess, monitor, and extend the lifetime of existing infrastructure.

RF7 Interfaces and interactions

RF3 and RF7 are complementary through their shared focus on reliability and risk assessment, while maintaining distinct scientific objectives. RF3 focuses on component reuse, material robustness, and sustainable construction materials, whereas RF7 focuses on structural lifetime extension, deterioration assessment, residual capacity estimation, and intervention planning for existing buildings and infrastructure. In this context, the structural condition assessment methodologies developed within RF7 provide an important basis for the reuse-oriented frameworks developed within RF3. The two fields therefore envision coordinated activities in reliability and risk methodologies, joint contributions to standardization and EC-related activities, and periodic knowledge-exchange meetings related to reuse, durability, deterioration, and lifetime extension strategies.

RF4 and RF7 share several enabling technologies, including AI, robotics, sensing systems, and digital twins, but apply them to distinct and complementary phases of the built environment lifecycle. RF4 focuses on digital fabrication, automated construction and deconstruction, interoperability, semantic modelling, and BIM-driven digital workflows, while RF7 focuses on structural health monitoring, condition assessment, deterioration quantification, residual capacity estimation, and risk-informed decision support for existing structures and infrastructure. The two fields therefore converge through clearly defined technical interfaces while maintaining distinct scientific ownership. In particular, RF4 develops enabling digital and automation ecosystems for the construction lifecycle, whereas RF7 develops engineering methodologies for interpreting structural condition, predicting deterioration, assessing reliability, and supporting lifetime extension decisions. To strengthen these interfaces, RF4 and RF7 envision co-hosted training and knowledge-exchange activities on topics such as agentic AI deployment, robotics platforms, digital twin engineering, physics-informed machine learning, and advanced sensing technologies.

RF5 and RF7 also interact through complementary scientific scopes. RF5 focuses on geotechnical systems, coastal systems, climate hazards, and environmental boundary conditions, whereas RF7 focuses on exposed structural systems of buildings and infrastructure. In many respects, the two research fields act as each other's boundary conditions, particularly in relation to soil-structure interaction, environmental loading, resilience, and climate adaptation. The two fields also share interfaces related to environmental and infrastructure monitoring, particularly in areas such as

remote sensing, satellite imaging, and soil-structure interaction assessment. Consequently, RF5 and RF7 envision joint activities related to interface problems and environmental boundary-condition assessment for resilient infrastructure systems.

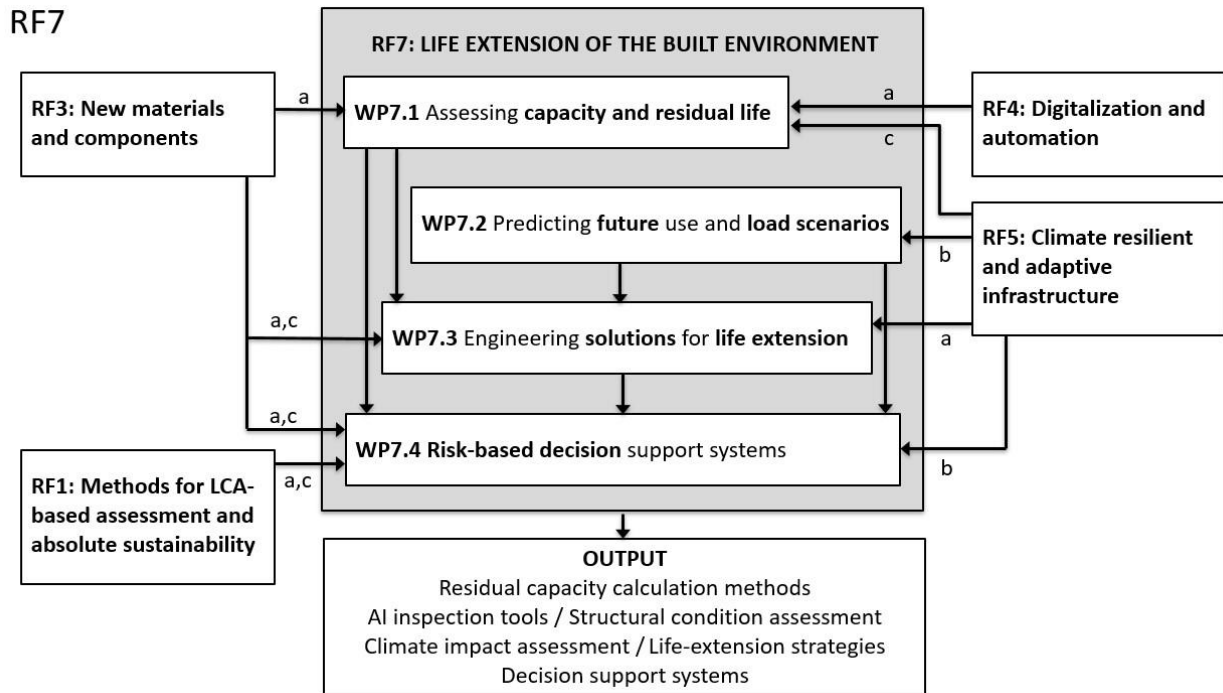


Figure 1. Interaction diagram

6. Deliver/receive necessary information from research results to enable research progress

4.6 Research Field specific output, outcome and impact targets

For the research field we have defined the following targets for which the realization hereof will be evaluated over the course of the programme.

Targets of the research field

Target name & description	Evaluation method	Impact Type ¹	Category ²	Years ³
Validated analytical and computational methods for capacity assessment of non-code-compliant RC structures	Publications	Output	New methods: Experimentally validated limit analysis and numerical (FELA and/or FESA) methods for calculating the capacity of non-code-compliant RC buildings and bridge structures	4

¹ Either *output*, *outcome* or *impact*.

² For *output*, it can be: *new method*, *new tool*, *new concept/approach*, *new technology*, *new material*, *other*. For *outcome*, it can be: *applied by industry*, *applied by consultancy*, *applied in regulation/standardization*, *applied in construction*, *demonstrated in practice*, *other*. For *impact* it can be: *change of practice*, *change of behaviour*, *change in perception*, *other*.

³ Which programme year (1=2026) the target expected is to be realized.

Experimental validated model linking corrosion to structural capacity	Publications	Output	New method: Experimentally validated corrosion models suitable for integration into limit analysis for the capacity assessment of corroded RC structures	4
Model predicting ASR effects on structural performance	Publications	Output	New method: Experimentally validated limit analysis approaches for evaluating the structural performance of RC structures affected by alkali-silica reaction (ASR).	4
Framework for capacity assessment and strengthening of RC buildings and bridges that are non-code-complaint and deteriorated by corrosion and/or ASR.	Publications	Output	New method: Experimentally validated framework that integrates in-situ testing with limit and/or numerical analysis to assess the structural performance of non-code-compliant structures affected by one or more deterioration mechanisms, including ASR and corrosion	8
Framework predicting fatigue crack growth in steel structures	Publications	Output	New concept/approach	4
Virtual sensing load estimation tool	Publications	Output	New tool: mathematical framework for evaluating applied loads based on measure structural response	4
Structural models integrating sensing and fracture mechanics	Publications	Output	New concept/approach	4
UAV based inspection system using computer vision	Publications	Output	New technology: description of cyber-physical system composed of UAVs, multi-modal sensors and computer vision to automate inspections on buildings and infrastructures.	4
UAV based inspection system using computer vision	Prototype	Output	New technology: proof-of-concept prototype of a cyber-physical system composed of UAVs, multi-modal sensors and computer vision to automate inspections on buildings and infrastructures.	10
AI/Robot assisted operation of structural health monitoring systems	Publications	Output	New technology: description of cyber-physical system composed of terrestrial robots, multi-modal sensors and orchestrated by AU to automate inspections on buildings and infrastructures.	4

AI/Robot assisted operation of structural health monitoring systems	Prototype	Output	New technology: proof-of-concept prototype of cyber-physical system composed of terrestrial robots, multi-modal sensors and orchestrated by AU to automate inspections on buildings and infrastructures.	8
Algorithm estimating wind loads from structural response data	Publications	Output	New tool: mathematical framework for estimation on wind-structure interaction models based on structural response data	4
Machine learning model predicting wind conditions for structures, with a focus on extremes	Publications	Output	New method: published SotA on evaluation methods for extreme wind and other environmental condition assessment	4
Enhanced machine learning model predicting wind conditions for structures, with a focus on extremes	Publications	Output	Advanced method: published enhanced methodology for extreme wind and other environmental condition assessment	10
Case studies defining reliability targets for existing structures	Publications	Output	New concept/approach: published risk-based assessment framework demonstrating reliability targets for multiple existing structures	4
Multi-physical model for wind action in the built environment	Publications	Output	New concept/approach	4
Consulting engineers apply new capacity assessment frameworks from initial projects	Survey and <i>fib</i> , IABSE, JCSS Bulletins	Outcome	Applied by consultancy: developed outputs are disseminated through <i>fib</i> , IABSE, and JCSS, enabling initial uptake and application by consulting engineers	4
Consulting engineers apply new capacity assessment frameworks from further developed output	Survey and <i>fib</i> , IABSE, JCSS Bulletins	Outcome	Applied by consultancy: following further development and dissemination, broader adoption of the methods by consulting engineers is expected and will be monitored through surveys	7
Consulting engineers apply new capacity assessment and strengthening frameworks that involve multiple of the outcomes developed	Survey and <i>fib</i> , IABSE, JCSS Bulletins	Outcome	Applied by consultancy: integrated frameworks combining monitoring, in-situ testing, and analytical and numerical models within a risk-based approach are disseminated through professional organizations and adopted by consulting engineers	10

Monitoring systems implemented for infrastructure asset portfolios	Published methodology	Outcome	Trialled by industry: engineering consultants test automated inspection systems based on robotics, AI and computer vision.	5
Enhanced monitoring systems implemented for infrastructure asset portfolios	Published enhanced methodology	Outcome	Applied by industry: infrastructure and building owners via engineering consultants adopts automated inspection systems based on robotics, AI and computer vision.	10
Wind loading estimation tool applied for retrofitting and lifetime extension	Published methodology	Outcome	Trialled by industry: engineering consultants test the wind loading estimation tool in the context of aerodynamic retrofitting and lifetime extension.	10
Enhanced wind loading estimation tool applied for retrofitting and lifetime extension	Published enhanced methodology	Outcome	Applied by industry: infrastructure and building owners via engineering consultants adopts the wind loading estimation tool in the context of aerodynamic retrofitting and lifetime extension.	10
Public agencies apply probabilistic assessment in maintenance planning	Survey	Outcome	Applied by industry: Public agencies (e.g., Danish Road Directorate and Railway Denmark) apply risk-based assessment methods for maintenance planning of infrastructure assets	10
Reliability concepts included in future design and assessment codes and recommendations	JCSS, fib, IABSE	Outcome	Applied in regulation/standardization: reliability indices derived from case studies are incorporated into bulletins and recommendations for the assessment of existing structures	6
Reliability concepts included in future design and assessment standards	CEN, Danish Standards or Danish road directorate bulletin and similar international docs	Outcome	Applied in regulation/standardization: reliability indices and risk-based assessment approaches are incorporated into standards, bulletins, and recommendations for assessing existing structures	10
Full-scale micro-pilot study applying developed monitoring and assessment tools	Practical case studies (requires additional funding)	Outcome	Demonstrated in practice: Micro pilot in full-scale conditions to assess new monitoring tools	4

Full-scale macro-pilot study applying developed monitoring and assessment tools	Practical case studies (requires additional funding)	Outcome	Demonstrated in practice: Micro pilot in full-scale conditions to assess enhanced monitoring tools	10
Impact assessment of climate change on wind load, structural response and durability	Publications	Outcome	Application in regulation/standardization	10
The service life of buildings and bridges are extended with least possible intervention	Survey and case studies	Impact	Change of practice: The developed assessment frameworks enable industry to extend the service life of existing buildings and bridges with least possible intervention.	10
Lower CO ₂ emissions from avoided reconstruction activities	Survey and case studies	Impact	Change of practice: the frameworks support improved decision-making that reduces CO ₂ emissions by enabling service life extension and avoiding unnecessary reconstruction.	10
Decision makers favour adaptation over demolition	Survey	Impact	Change in perception: reliable assessment methods enable decision-makers to prioritize adaptation and extended service life over demolition of existing structures	10

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